

IoT Enabled Bee Hive Monitoring System

M. Milovanović, J. Pejić, and P. Pejić

Abstract - Beehive monitoring can provide useful information for people associated with beekeeping to help to manage their honey bee (*Apis mellifera*) colonies. This paper presents the development and experimental results of a IoT enabled bee hive monitoring system

I. INTRODUCTION

The fact that bees pollinate about 85% of flowers, make them as one of the most important insects. Because of the importance, there is an increased interest in bee lives and behavior. Precision Apiculture or Precision Beekeeping (PB) is recently defined scientific direction which combines Information Technologies (IT) and beekeeping. PB is defined as an apiary management strategy based on the monitoring of individual bee colonies to minimize resource consumption and maximize the productivity of bees.

Problem of the precision beekeeping has already been tackled before [1-4]. Zacepins described that data collection process in PB can be classified into three groups [1]. First group are apiary-level parameters (meteorological parameters and video observation), second group are colony-level parameters (temperature, humidity, gas content, sound, video, vibration of hive and weight) and the third group are individual bee-related parameters (the number of incoming/outgoing bees, the number of bees in the hive entrance area. Authors provide detailed information about challenges, development perspectives and need for all three types of this measurement [1].

Measurements of apiary-located meteorological parameters, and individual bee-related parameters may explain some peculiarities in colony behavior. This type of information can increase the detailed understanding of colony-level measurements. Bee colonies can be automatically monitored using different approaches and methods [1-5]. In system architectures sensor's data gathering can vary – data transmission (wired or wireless technologies), data storage and usage of intermediate devices. A particular monitoring system can also be supplemented with decision support system, which is

responsible for data analysis, bee colony condition determination and appropriate decision selection. Kviesis described six approaches to developed a beehive monitoring system [2]. Approaches also include on-site and remote data monitoring, and how data from bee colony can be transferred to the beekeeper for real-time data monitoring. In this experiment in this experiment, the temperature inside the hive was measured, since the temperature is the factor that can be easily monitored and it is economically feasible.

Zacepins describes complex bee colony temperature measurement system [3]. Developed multi-node temperature measurement system for bee colony online monitoring consists of two parts: remote measurement node for real time bee colony temperature measurements and web server for measurement summarizing and demonstration to the end users.

Zabasta investigated the quality and reliability of WSN network, which is used for bee hives monitoring system [4]. This research explores the problems associated with vulnerability of Wireless Sensor Network, with the special focus on the weather conditions impact on the signal transmitting quality in WSN. The study of impact of weather conditions on the signal strength in outdoor wireless sensor networks, testing reliability of data transition, alerting a service performance degradation, the data transmitting joint concept in different weather conditions, based on the WSN testing results, was implemented in this research.

The goal of this paper is to describe low-cost system for precision beekeeping that collects temperature, humidity and the weight of the beehive based on IoT. By monitoring the temperature, humidity and the weight of Bee hive it enables bee-keepers to get real-time information from hives and partners in biology to study pests and diseases. This kind of system will significantly reduce the number of beehive inspections by beekeepers, which will increase labor productivity and rapidity of response on emergency situations.

II. SYSTEM DESIGN

There are six functional units that make up this system: (1) the central computer unit, (2) unit for communication, (3) the weight monitoring unit, (4) unit for measuring external temperature and humidity, (5) GPS and (6) power supply. The block diagram of the system is shown in Fig. 1. The whole system, except power supply unit is placed in two-part case placed beneath the beehive.

M. Milovanović, is with the Department of Electronics, Faculty of Electronic Engineering, University of Niš, Aleksandra Medvedeva 14, 18000 Niš, Serbia, E-mail: mihajlo.milovanovic@elfak.rs

J. Pejić is with the Department of Computer Science, Faculty of Sciences and Mathematics University of Niš, Višegradska 33, 18000 Niš, Serbia, E-mail: jelena.milovanovic@pmf.edu.rs

P. Pejić is with the Faculty of Information Technology, Belgrade Metropolitan University, E-mail: petar.pejic@metropolitan.ac.rs

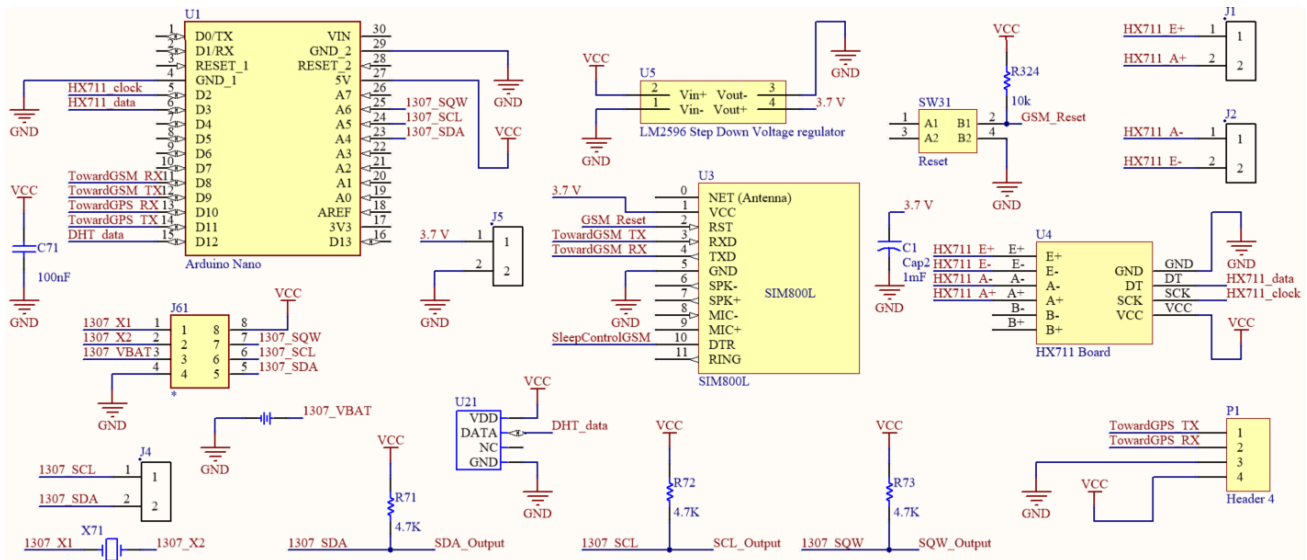


Fig. 1. Block diagram of the designed system.

Important functional part of the hardware is the power supply. The system is powered by a photovoltaic system as beekeepers places hives in rural areas where standard power supply is not available and during a foraging season, beekeepers are forced to reallocate bee colonies several times. Power unit consists of the standard components: solar module, charging controller and battery.

For finding a location of the hive, NEO-6M GPS module is used [6]. Its compact architecture, power and memory options make this module ideal for battery operated mobile devices with very strict cost and space constraints such as this one [7]. Also, external antenna is used, because it is desirable to keep the antenna at a distance from the board. The main reason why GPS is used more and more in digital beekeeping is protection against theft [1, 3]. By using a GPS tracking device, beekeepers can increase visibility, improve accountability, and safeguard beehives and equipment from thieves.

Temperature is the factor that can be easily monitored and it is economically feasible. Temperature monitoring can be completed using various information technologies, systems and methods. In our system DHT22 module is used, digital-output relative humidity and temperature sensor [8]. Small size, low consumption and long transmission distance make DHT22 perfect for this purpose.

The weight monitoring system is based on four identical printed capacitive force sensors placed on the bottom board (floor) of the beehive. An arrangement of four 50 kg load cells is used in conjunction with a load cell transmitter HX711 [9]. It is well known by the experience of beekeepers that the weight of hive oscillates from roughly 10kg to 200 kg. That's the main reason for use 4 load cells.

Transmitting module which is responsible for sending the data from each hive is a miniature SIM800L

GSM/GPRS module [10]. At the heart of this module is a SIM800L GSM cellular chip from Simcom. The operating voltage of the chip ranges from 3.4V to 4.4V. The module supports baud rates ranging from 1200 bps to 115200 bps. because it is desirable to keep the antenna away from the board again. 3dBi GSM antenna with a U.FL to SMA adapter is used. This type of antenna also provides better performance.

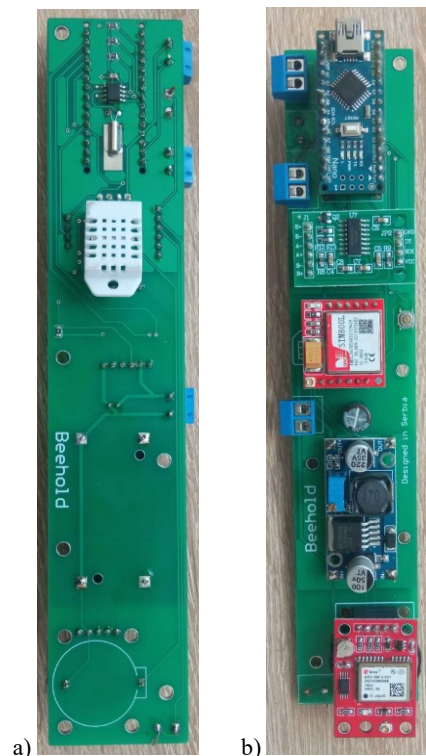


Fig. 2. a) Top and b) bottom side of PCB.

The disadvantage of SIM800L is that it can be a relatively power-hungry device, depending on its state. The

module's maximum current draw is around 2A (especially during a transmission burst). Because of this, when data is not transmitting, SIM800L module goes to the sleep mode, so that the current consumption is as low as 0.7mA. Operating voltage of this module is 3.4 up to 4.4V and the power supply of the rest of the system is 5V. To overcome this problem a 2A-rated DC-DC buck converter, named LM2596 is used. Requiring a minimum number of external components, this type of buck regulators is simple to use.

In order for the collected data of the hive to be relevant, one of the most important elements is time precision. To provide this timing, and to add real time clock functionality, component DS1307 chip is used [12]. In order to ensure the constant accuracy of the time, even in the event of an unforeseen temporary power outage, an additional 2032 Li-ion battery is used.

After successful recording, the data is transferred through mobile network. The data is sent to the remote server which forms information messages and reports for the web applications on the beekeepers' mobile devices. Connection to the server and data transfer to the database is performed using SIM800L, controlled with AT commands. Each developed system has its own ID (identifier) so that it can be uniquely assigned on the data base [13, 14].

ATmega328p that is mounted onto the Arduino nano board is used as the central processing unit responsible for the functioning of the entire system [15]. The firmware is developed using the Arduino IDE, with the C programming language.

III. TESTING OF THE SYSTEM

The system is designed so that it can work in four modes depending on the beekeeper's choice. The first mode collects and sends all data once a day. At 09:00 and is suitable for the winter period of the year due to its low consumption and the decreased mobility of the bees. The second mode collects data five times a day and send only once, at 09:00. The reason for the existence of this mode is because beekeepers sometimes do not need real-time monitoring, but to see a report at the end of the day. The third mode is very similar to the second, with the difference that it sends reports twice, at noon and at the end of the day. The fourth mode was found to be the most frequently used by beekeepers. In this mode, the device collects and sends data five times a day, at 05:00, 09:00, 13:00, 17:00 and 21:00.

Figure 3 shows the data collected from the device that is in fourth mode. Five readings on May 17, 2023. The figure shows that the weight of the hive remains pretty much the same, while humidity and temperature changes during the different phases of the day.

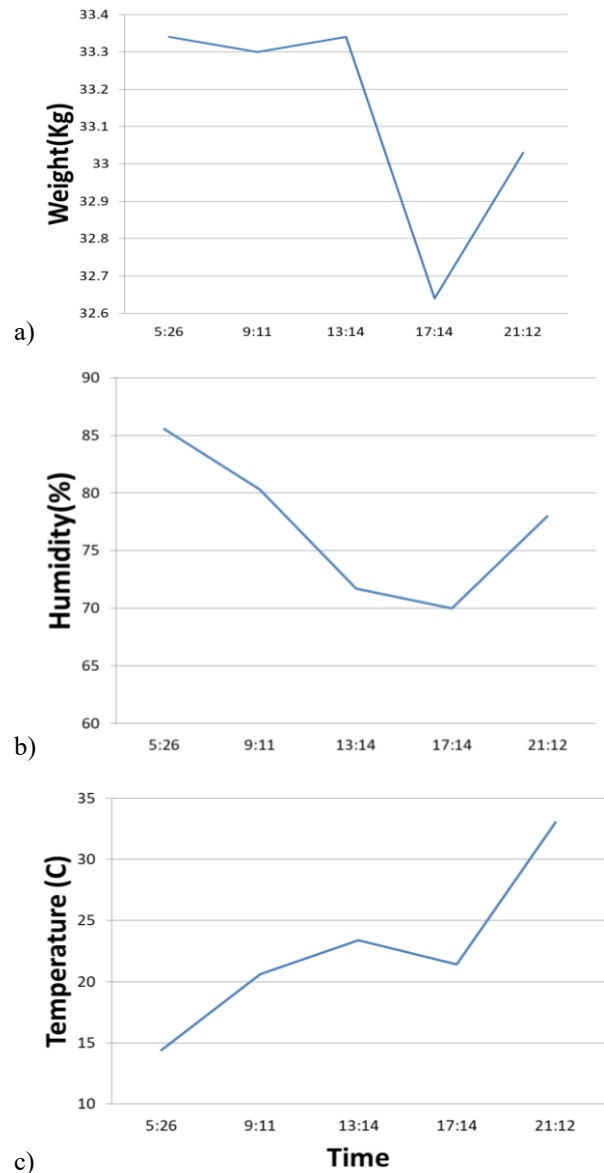


Fig. 3. Experimental results of beehive parameters obtained during 16 hours period: a) weight; b) humidity; c) temperature.

Based on the received parameters beekeepers can see the change in the weight during the day, but also assess whether the beehive is in danger of freezing or overheating during some phases of the day.

Still, one-day results, offer limited basis for the assessment of the position and conditions for the beehive. For the proper evaluation, it is needed to perform measurements and gather data for the longer period of time. Figure 4. shows the data collected from the device that is in first mode, where the data read occurs every evening at 21:00 for a period of one month.

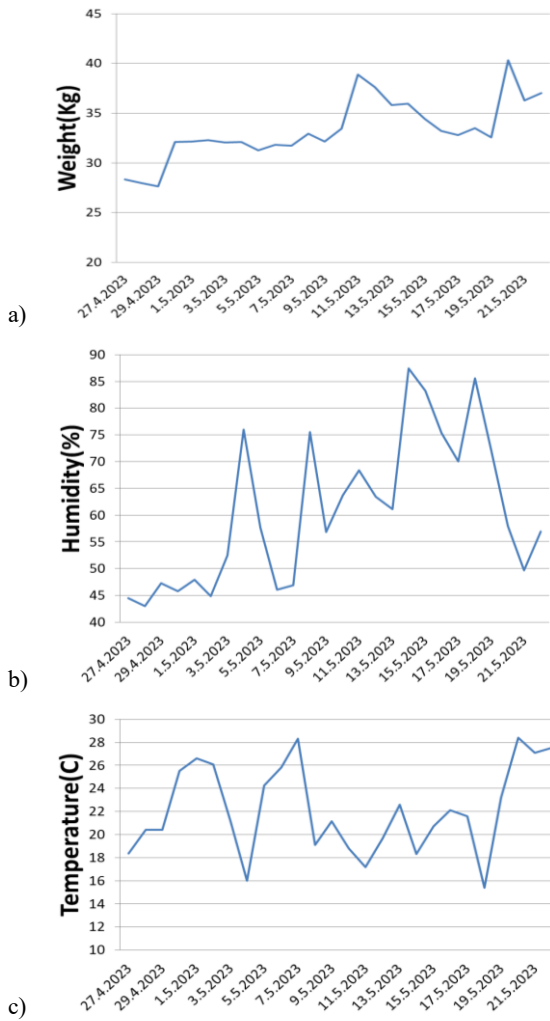


Fig. 4. Experimental results of beehive parameters obtained during one month period (30 measurements): a) weight; b) humidity; c) temperature.

With the provided results, several notes should be taken. Weight of the beehive increases with time, because bees create more honey. This is expected, but, in the Fig. 4, some spikes can be seen. During the regular maintenance, load cells used for measuring weight should be somehow disabled or not influenced by the beekeeper. Spikes that occur due to the physical shift of the beehive that disrupts reliable weight measuring. On the other hand, temperature and humidity values do not exceed some critical values that drastically impact the honey generation process. Since the measurements were done in the spring, it is yet to be analyzed how the exceedance of some critical values of temperature and humidity should be tackled by the beekeepers.

IV. CONCLUSION

At this moment, several devices are in the field. The implemented integrated system collects the most important

data from the hive, performs its primary processing and transmits it to a web server. Beekeeper is able to monitor apiary status online and react only if necessary.

Also, in some parts of the country where the described systems are installed, the weather has been quite variable. With this kind of testing, it is proved that some of the weather conditions, such as rain or a sudden change in temperature, do not significantly affect its operation.

ACKNOWLEDGEMENT

The authors would like to thank to the company Beehold for the generous support and aid in development and testing of the described research.

REFERENCES

- [1] A. Zacepins, V. Brusbardis, J. Meitalovs, and E. Stalidzans, "Challenges in the development of Precision Beekeeping", *Biosystems Engineering*, 2015, vol. 130, pp. 60-71
- [2] A. Kviesis and A. Zacepins, "System Architectures for Real-time Bee Colony Temperature Monitoring", *Procedia Computer Science*, 2015, vol. 43, pp. 86-94.
- [3] A. Zacepins and J. Meitalovs, "Implementation of multi-node temperature measurement system for bee colonies online monitoring", *Proc. of the 2014 15th International Carpathian Control Conference (ICCC)*, 2014, pp. 695-698.
- [4] A. Zabasta, A. Zhiravetska, N. Kunicina, and K. Kondratjevs, "Technical Implementation of IoT Concept for Bee Colony Monitoring", *Proc. of 2019 8th Mediterranean Conference on Embedded Computing (MECO)*, 2019, pp. 1-4.
- [5] M. Milovanović, P. Pejić, and J. Pejić, "Sensor detection of honeycomb detection", *Proc. of International Conference of Experimental and Numerical Investigations and New Technologies*, 2021
- [6] NEO-6M module datasheet, U-blox, 2011.
- [7] N. Mitrović, S. Veljković, Z. Prijić, and D. Danković, "Comparison of the performance of the different GPS receivers in practical applications", *Proc. of IEEE Zooming Innovation in Consumer Technologies Conference (ZINC)*, Novi Sad, Serbia, 2022, pp. 11-16.
- [8] DHT22 datasheet, Aosong Electronics Co., 2018.
- [9] HX711 datasheet, Avia Electronics, 2011.
- [10] SIM800l datasheet, SIMcom, 2013.
- [11] A. Biswas, A. Mondal, and P. Sarkar, "Design and Implementation of Digital Controller in Delta Domain for Buck Converter", *Facta Universitatis, Series: Electronics and Energetics*, 2023, vol. 36, no. 1, pp. 103-119.
- [11] DS1307 datasheet, Maxim Integrated, 2015.
- [12] N. Mitrović, M. Đorđević, S. Veljković, and D. Danković, "Implementation and Testing of WebSocket Protocol in ESP32 based IoT systems", *Facta Universitatis, Series: Electronics and Energetics*, 2023, vol. 36, no. 2, pp. 267-284.
- [13] N. Mitrović, M. Đorđević, S. Veljković, and D. Danković, "Implementation of WebSockets in ESP32 based IoT Systems", *Proc. of 15th International Conference on Advanced Technologies, Systems and Services in Telecommunications (TELSIKS 2021)*, Niš, Serbia, 2021, pp. 261-264.
- [14] ATmega328P datasheet, Atmel, 2015.